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Greening IMF Quotas: A Proposal for Paris Alignment

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Finance Department

Greening IMF Quotas

Prepared by Flavien Moreau, Mengxue Wang, Shan He*

Authorized for distribution by Olaf Unteroberdoerster/Zuzana/Heikki/Greetje
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ABSTRACT: Is the current way of deriving the IMF members' quotas well aligned with the increasing challenges due to climate change? This paper analyses the potential impact of climate change and of climate mitigation strategies on the variables that enter the quota formula. It also considers the case for aligning the formula with global carbon neutrality commitments. We provide simulations of what such a re-alignment could look like under multiple climate-related scenarios, and find that a simple and transparent adjustment could deliver such a re-alignment with minimal abrupt changes while protecting the poorest members.

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WORKING PAPERS

Greening IMF Quotas: A Proposal for Paris Alignment

Prepared by Flavien Moreau, Mengxue Wang, Shan He[1]

[1] The authors would like to thank Olaf Unteroberdoerster for guidance and support, as well as Zuzana Murgasova, Heikki Hatanpaa, Alberto Sanchez Rodelgo, Thomas Stoerk, Florent McIsaac, and IMF departmental reviewers for their helpful advice and comments. The views expressed in this Working Paper are those of the authors and do not necessarily represent those of the IMF, IMF Board, or of the Finance Department. An initial version of the proposal was submitted to the IMF Climate Change Innovation Challenge in 2020.

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I. Why Greening Quotas?

Quotas play a central role in the Fund, including by capturing member countries' capacity to contribute resources to the Fund and serving as a guide to regulate access to Fund resources. In particular, quota resources are currently the sole immediately available financing source for new Fund lending, with borrowed resources providing a critical backstop to quotas in case of larger needs. Quotas, together with other metrics, are also used in the Fund's multi-layer risk management approach in order to assess member countries' capacity to repay the Fund and mitigate credit risk. Finally, quotas also determine members' voting rights and members' shares in a general allocation of Special Drawing Rights (SDRs).[1]

Climate change is expected to impact both countries' need for resources and ability to contribute, which could exacerbate the tension between quotas' multiple roles. While climate change will likely leave most countries worse-off, it is expected to hit poorer and vulnerable countries the hardest. Coincidentally, low-income countries tend to be the countries with the largest access need relative to their economic size, while, at the same time, having a lower capacity to contribute. The need for additional climate financing for vulnerable countries is one key motivation behind the recently created Resilience and Sustainability Trust Fund.[2]

Substantial damages from climate change call for the implementation of immediate, global, and multi-pronged efforts towards a deep decarbonization. Especially concerning, global GHG emissions have rebounded in 2021, after a short-lived 4.6 percent decline in 2020 following the Covid-19 pandemic.[3] Immediate and deep emissions reductions concerted among countries are required to limit global warming to 1.5°C, according to the 2021 IPCC report.[4] Climate change as well as mitigation and adaptation policies will profoundly affect several of the key economic variables, including GDP, trade, and reserves, on which the calculation of Fund's quotas is based. This poses important challenges for the future of the Fund's governance and alignment with its core mandate.

In their current form, Fund quotas do not explicitly recognize climate externalities from GHG emissions, therefore implicitly rewarding carbon-emitting economic activities. As a result, the incentives embedded in the current formula are at odds with the objectives of promoting sustained and green development, enshrined in the Paris Agreements. In addition, many countries hit the hardest by the COVID-19 pandemic have not fully recovered and are facing complex tradeoffs between carbon-fueled catch-up growth and more environmental-friendly development. The current quota formula is not aligned with the more climate-friendly options and can therefore be described as "carbon-blind". As a result, it is not helping align interests towards international coordination on a global sustainable path.

[1] As determined by the Articles of Agreement (Article XVIII, Section 2(b)), the distribution of the SDR allocation is made in proportion to quota shares. A general allocation of SDRs equivalent to US$650 billion (about SDR 456 billion) was approved by the Board of Governors of the IMF on August 2, 2021, to boost global liquidity. See *Allocation of Special Drawing Rights for The Eleventh Basic Period: Draft Executive Board Decision and Managing Director Report to the Board of Governors*. IMF Policy Paper No. 2021/050.

[2] See *Proposal to Establish a Resilience and Sustainability Trust*. IMF Policy Paper No. 2022/013.

[3] Annual Greenhouse Gas (GHG) Air Emissions Accounts, IMF Climate Change Indicators Dashboard. https://climatedata.imf.org/datasets/c8579761f19740dfbe4418b205654ddf/explore. According to the IPCC Summary for Policymakers Headline Statements on April 4, 2022, "global GHG emissions are projected to peak between 2020 and at the latest before 2025 in global modeled pathways that limit warming to 1.5°C..."

[4] Masson-Delmotte et al. 2021.

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Simple amendments to the quota formula could serve as a guide to better align with the Paris agreements the need for strong Fund governance reflecting members' standing in the global economy. The Articles of Agreement set out the key roles for quotas, but do not indicate how members' quotas should be determined. In practice, quotas have been related to, but not strictly determined by, economic factors. Therefore, we consider "greening" the quota formula and propose a simple, operational adjustment that would preserve the current formula's structure, including the variables and weights that were the result of a large consensus across the membership, while "greening" the variable themselves. Instrumental in this greening of the quota formula, a social cost of carbon emission would be used to account for negative climate externalities in the GDP variable and could be flexibly adjusted over time or across country groupings.

A green quota formula could serve as a useful commitment device to help coordinate climate ambitions and foster Fund members' climate mitigation efforts. By explicitly acknowledging both the negative externalities posed by carbon-emitting economic activities and the international commitments towards net zero emissions, such a formula would provide an additional reward to countries reaching their GHG emission reduction targets. In other words, it would help ensure that the Fund's governance is incentive-compatible with global commitments towards deep decarbonization of the economy. This incentive-compatible governance could also help set the Fund on stronger foundations and cement climate commitments in the face of increasing damages and uncertainty caused by climate change.

By building on the existing formula the approach would preserve several characteristics that define a good quota formula candidate. It should (i) be simple and transparent, so that the basis for differences in relative quota shares is readily understandable, (ii) be consistent with the multiple roles of quotas, (iii) produce results that are broadly acceptable to the membership, and (iv) be feasible to implement statistically based on timely, high quality and widely available data.[5]

In the following sections, we will first propose our approach to greening Fund Quotas and what it could imply to the current quota system. Then we will show the development of quota share and green quota share using the current quota formula under different established climate scenarios. Finally, we will lay out directions for future research at the juncture of IMF quotas and climate changes.

II. A Simple and Practical Approach to Greening Quotas

Potential methods to greening Fund quotas should satisfy at least two conditions: availability of reliable data for all Fund members and timeliness. First, Calculated Quota Share (CQS) is supposed to be calculated for every member country of the IMF. The data sources that are used to calculate variables should be available to cover all IMF member countries. Some advanced economies calculate carbon-neutral macro variables which would otherwise be ideal for green quota calculation if not for their limited availability. Secondly, the adjusted variables should be easy to calculate and accessible to ensure timely updates of quotas. Variables that are calculated and published with a multi-year delay are thus not a good candidate.

[5] See *Fifteenth General Review of Quotas—Quota Formula and Realigning Shares*. IMF. Policy Paper No. 2021/007.

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A flexible approach preserving essential features on the current formula would also have merit. Both the actual quotas and the quota formula have proved resilient over time, underscoring the need for a well thought-out and robust quota formula. The current formula was agreed in 2008. And the current quota shares have remained unchanged since the implementation of the 14th review. A forward-looking approach which, in this case, is the formula that could consider the impact of climate changes on quota formula, can therefore reduce the risks of misalignments being locked in for an extended period of time.

We consider a simple approach to greening Fund quotas that we illustrate by focusing on the main component of the quota variable, GDP. The approach preserves the current formula and focuses on greening the existing variables. This approach recognizes that quota variables are "gross" concepts and proposes green corrections to these variables. By definition, these gross variables abstract from environmental externalities are at best imperfectly priced (e.g., *Gross* Domestic Product). However, in some cases, there are natural ways to correct them, for instance by using a price of carbon.[6] We will show the detailed way of doing so in this sector.

A. Existing Quota variables

The current quota formula consists of four key macro variables, namely GDP, openness, variability and reserves. Box 1 explains the detailed formula calculation and variable definitions. We discuss the relevancy and potential methods of greening each of these four variables.

GDP: GDP takes up the largest share in the quota formula and there is broad support among the membership for GDP to remain the most important variable.[7] We thus focus on GDP as is our main variable for discussion. Greening the GDP variable is highly relevant for calculating quota shares and is also attracting attention from the academic community on proper ways of doing so. We will discuss in detail in later context on how it is greened.

Openness: Openness mainly measures the trade component of member countries. There is also a large literature discussing how to measure green trade[8]. However, due to data limitations for most of the IMF member countries, this paper does not aim at greening the openness variable. But we leave it as an important field for future research to provide a better measurement of "green quota."

Variability: Variability measures the volatility of the openness variable. Since the discussion of greening openness is already out of the scope of this paper, we leave variability for future research.

Reserves: The most common reserve assets are treasury bills, currencies, gold, or SDR. While there is a burgeoning activity in various forms of "green" or "blue bonds" that take environmental concerns into consideration, it is difficult at the current juncture to provide a comprehensive taxonomy of green reserves and we leave this work for future research.[9]

[6] An alternative approach would be to consider building a hybrid formula, integrating broader environmental externalities that have no obvious price, e.g., biodiversity. The construction of hybrid indicators of economic and social welfare has been an active research area since the 1990s, see Fleurbaey (2009) and EU Commission Staff Working Document (SWD(2013) 303) on "Progress on 'GDP and beyond' actions".

[7] See *Sixteenth General Review of Quotas - Further Considerations* (EB/CQUOTA/22/1), paragraph 17.

[8] See for example the literature discussion on cross-border carbon tax (Kang and Lee 2021).

[9] For an overview of green debt instruments, see Green Bonds (worldbank.org).

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Box 1: The Current Quota Formula

The current quota formula was agreed in 2008. It includes four variables (GDP, openness, variability, and reserves), expressed in shares of global totals, with the variables assigned weights totaling to 1.0. The formula also includes a compression factor that reduces dispersion in calculated quota shares (CQS). The formula is:

$$CQS = (0.5 \times Y + 0.3 \times O + 0.15 \times V + 0.05 \times R)^k$$

where the variables are defined as follows,

CQS = calculated quota share;

Y = a blend of GDP converted at market exchange rates and PPP rates averaged over a three-year period. The current weights of market-based and PPP GDP are 0.60 and 0.40, respectively;

O = the annual average of the sum of current payments and current receipts (goods, services, income, and transfers) for a five-year period;

V = variability of current receipts and net capital flows (measured as the standard deviation from a centered three-year trend over a thirteen-year period);

R = twelve-month average over one year of official reserves (foreign exchange, SDR holdings, reserve position in the Fund, and monetary gold); and

k = a compression factor of 0.95. The compression factor is applied to the uncompressed calculates quota shares which are then rescaled to sum to 100.

B. Greening GDP

A theoretically appealing solution is to construct a broad measure of "Green" GDP that nets out all the environmental externalities generated by economic activities while integrating environmental services. This approach would be in line with a large literature aiming to correct the well-known limitations of the GDP.[10] There are different methods of calculating such a "Green" GDP, each meant to address a specific externality. For instance, the green GDP indicator by Stjepanović et al. 2017 is calculated as a traditional GDP indicator minus the cost of CO_2 emissions, the cost of natural resource consumption, and the cost of environmental depletion. It also integrates supplementary information by distinguishing the real costs of environmental damage and opportunity costs of a lost turnover.[11] Attempts to implement these approaches have been conducted at both national and provincial levels, and tend to require specific and detailed information, not

[10] For an example of these critiques, see Stiglitz et al. 2018: "While GDP is the most well-known, and most powerful economic indicator, it can't tell us everything we need to know about the health of countries and societies. In fact, it can't even tell us everything we need to know about economic performance."

[11] The exact formula used by Stepanovic is as follows:

$$Green\ GDP = GDP - (CO_2\ emissions\ in\ thousand\ metric\ tons \times carbon\ price\ per\ thousand\ metric\ ton) - (metric\ ton\ of\ waste \times 74\ kWh\ of\ electrical\ energy\ per\ metric\ ton \times price\ for\ 1kWh\ of\ electrical\ energy) - (GNI/100 \times natural\ resources\ depletion\ in\ \%\ of\ GNI)$$

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always widely available.[12] Stjepanović et al. 2017 performed a cross-country analysis including the depletion of natural resources and degradation of the environment in calculating GDP, but only on a restricted sample of 44 countries.

A more transparent and simple option is to construct a measure of "Green" measure of GDP that simply substract the externality created by carbon emissions using a suitable carbon price. The formula can therefore be written as follows:

$$Green\ GDP = GDP - CO_2\ emissions\ in\ thousand\ metric\ tons \times carbon\ price\ per\ thousand\ metric\ ton$$

While leaving aside some environmental externalities, this approach would rely on data currently available across a good coverage of IMF membership (91 percent of IMF membership to be precise, see footnote 2 of Figure 2 for more information) and can potentially cover the whole membership in the future that would satisfy quota data quality standards. Furthermore, as the cost of natural resource consumption and the cost of environmental depletion are mostly local externalities, it is reasonable to assume that, under the quota setting, we are more interested in the global externality from GDP, which is the cost of CO_2 emissions. Pricing carbon externalities directly in the existing quota variable allows for a simple, flexible, and replicable method, with strong foundations in the literature[13].

This solution is also more practical than alternative modification of the quota formula, such as adding a fifth, specifically green variable.[14] Adding such a variable would raise at least three difficulties: i) a necessary reweighting of the other variables with complex discussions on the appropriate weights, ii) the need to reconcile a non-monetary quantity (GHG emissions) with the other four monetary variables, and iii) the need to find an appropriate method to make shares in GHG emissions into quantities count negatively towards quota shares. While not insurmountable in isolation, each of these challenges constitutes potentially lengthy methodological detours.

The choice of an appropriate carbon pricing is a key component of the proposed approach to greening GDP. A first option is to choose a uniform carbon price across countries. Under this assumption, tons of carbon emitted would be priced exactly the same regardless of levels of economic development. One possible justification for this assumption would be a sole concern with the global climate externality generated by carbon emissions: if we regard reducing carbon emissions as generating a global public good, then the same good should be priced the same for all countries providing this good, i.e., reducing emissions.

There are good reasons to price carbon emissions differently across countries at different levels of development, once broader welfare and fairness implications are considered. First, countries with higher carbon intensity of GDP have more room to cut carbon emissions compared with countries with low carbon

[12] For instance, Yu et al. 2019 examined the prefectural level ecosystem service value in China by accounting for the land use and crop output from 2009 to 2015 to aggregate the green GDP in China. Liang et al. 2021 measured the gross ecosystem product of nine cities within the Pearl River Delta of China by detailed examining products and services that the ecosystem provided for human welfare and economic and social sustainable development.

[13] See for example Fleurbaoy 2009

[14] Such an approach – combining economic and non-economic variables in a single index -- would be reminiscent of earlier index-based measures as the Human Development Index (HDI), whose composition is nevertheless dominated by GDP.

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intensities.[15] These considerations suggest that higher carbon prices for these countries with a higher carbon price elasticity may result in more emission reduction. Second, imposing a lower cost of carbon for less-developed countries partly addresses the potential needs of quota adjustments to protect the quota shares of poorer Fund members and help keep the quota formula parsimonious. Lastly, the use of differentiated CO2 prices is parallel to the use of market GDP, which relies on local prices of goods and services. We experimented with differentiated carbon prices, which is shown in footnote 2 of Figure 2, supposing the countries and regions set prices differently to achieve carbon emission goals, and the results are shown in Figure 2.

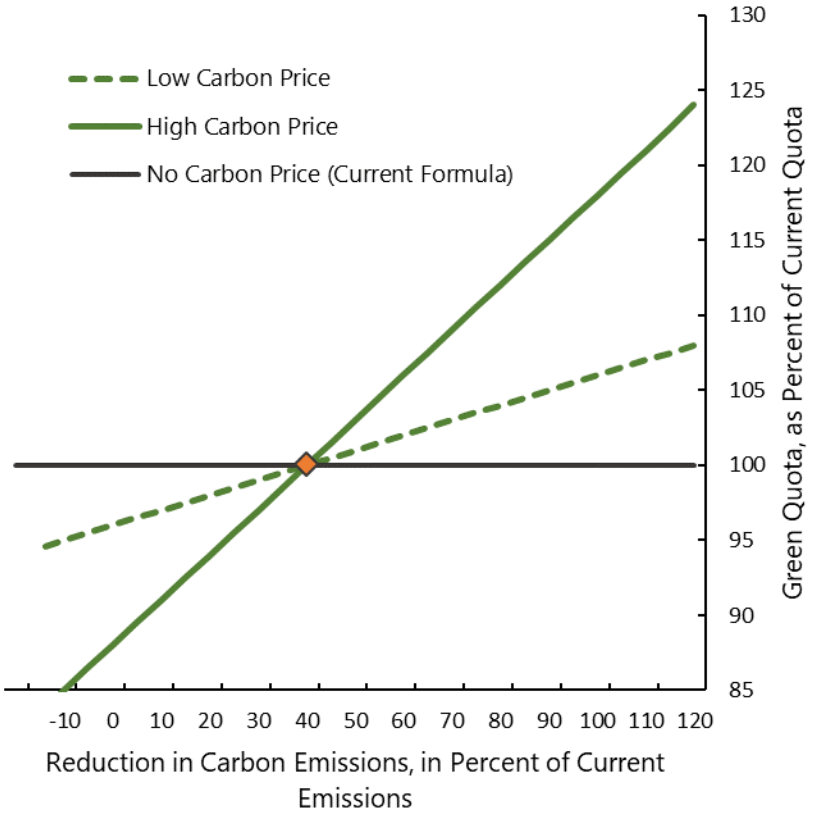
Another important decision is the use of gross as opposed to net emissions. Net emissions refer to CO_2 emitted by the country net of the CO_2 taken out of the atmosphere. From a theoretical standpoint, net emissions should be subtracted from GDP in the calculation instead of gross emissions, as the overall burden to the environment is the *net* emissions that could not be neutralized. However, for the purpose of the illustrative calculations on the current shares in Figure 2, we use gross emissions, mainly for two reasons. First, we are not aware of a publicly available dataset on net emissions. Second, at the current juncture, there are only pilot carbon capture and storage (CCS) projects and therefore negligible negative emissions. This is one caveat of the calculation and future works can focus more on constructing such a net emission database.

A Green GDP constructed along those lines would be more incentive-compatible with climate mitigation efforts, in the sense that efforts to reduce the carbon intensity of the economy would translate, ceteris paribus, in larger quota and hence a stronger voice at the Fund. To see this, consider an illustrative relationship between green quota and carbon emission reductions (Figure 1). In the current case, carbon emissions are effectively not counted, equivalent to a price of zero. Therefore, the quota share is indifferent to emission reductions. In contrast, a positive price on emissions creates a positive relationship between quota shares and emissions reduction.

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[15] See *Proposal for an International Carbon Price Floor Among Large Emitters*. IMF Staff Climate Note No 2021/001.

Figure 1. Linking quotas to carbon emission reductions with a Green GDP [1/]



Source: IMF staff calculation.

1/ Under the current framework, the GDP variable entering the quota formula is indifferent to carbon intensity (solid black line). In contrast, with a carbon price in the quota formula creates an explicit positive relationship between carbon emissions reductions and quota shares. A higher carbon price would increase the slope at which Green GDP and GDP diverge depending on carbon intensity. For any given country, reaching net zero emissions would equalize Green GDP and GDP. The figure assumes that other countries' emissions are unchanged.

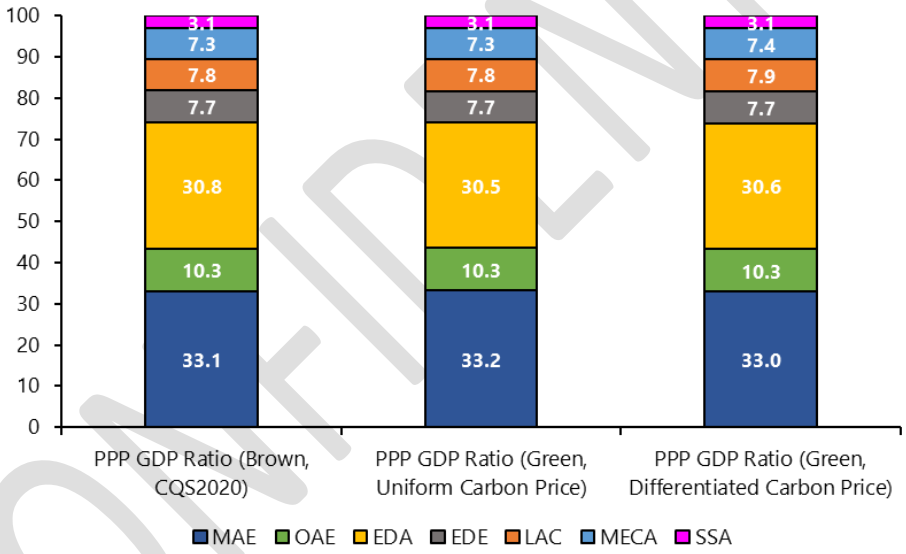
C. A simple application on the current Quota data

Introducing a Green GDP in the quota formula with a moderate price of carbon would not dramatically alter the current composition of quota shares among major country groupings. With a uniform price of carbon at around $75/tonne, which is in the middle of the range considered in the literature (see Stiglitz et al. 2017) and in line with previous Fund policy (Parry et al. 2021, IMF Staff Climate Notes 2021/001), the current quota share allocation across country groupings would be preserved. This reflects the fact that with a modest carbon price of $75, global carbon emissions would be priced at around $3Tr, or 3.5[16] percent of global GDP. Naturally, choosing a higher price of carbon more in line with the full social cost would magnify the impact of the greening of the formula.

[16] The World Bank estimates global GDP of 2020 to be 84.71Tr USD.

As we can see in Figure 2, a Green GDP formula might not necessarily result in large rebalancing across country groupings. There are large variations in carbon intensity per unit of GDP both within and across country groupings, with a rebalancing of a few decimal points towards Major Advanced Economies.[17] A price scale can be used to value carbon emissions for both development and quota purposes. In the chart below, we also experimented using differentiated price levels other than the uniform price of carbon at $75/tonne, following the arguments laid out earlier in the section. With differentiated carbon price, shares are indeed shifted towards countries imposed with lower carbon price, i.e. developing and poorer countries, compared to the calculation with uniform carbon prices. Additionally, the calculation of green shares in consideration of the cost of natural resource consumption and the cost of environmental depletion supports our findings about modest rebalancing (See Figure II.1. in Annex II).

Figure 2. Comparison of Current Brown and Green Shares of PPP GDP by WEO Classification (in percent) 1/ 2/



Source: IMF Finance Department; World Development Indicators database (updated by June 30, 2022), World Bank; and IMF staff calculation.

1/ Countries are classified based on WEO October 2021 groupings. MAE: Major Advanced Economies, OAE: Other Advanced Economies, EDA: , EDE, LAC: Latin America and the Caribbean, MECA: Middle East and Central Asia, SSA: Sub-Saharan Africa.

2/ PPP GDP data are from the Quota Data Update 2020 database. The Green PPP GDP indicator is then calculated based on the formula specified in section II.B, by subtracting the cost of CO_2 emissions. The CO_2 emissions (from the burning of fossil fuels and the manufacture of cement) and gross national income (expressed in PPP term) were retrieved from World Bank's World Development Indicators database. We assumed the carbon price as 75 USD per tCO_2 for all countries for the three years in calculation (2016-2018), considering the carbon price level as at least 40-80 USD per tCO_2 by 2020 in Stiglitz et al. 2017 and the annual price rise of 1.5 USD prior to 2020 assumption in Parry et al. 2021 (IMF Working Paper WP/21/236), consistent with the Paris temperature target limiting global warming to well below 2°C. We also experimented with differentiated carbon price for different country groups, 75 for MAE and OAE, 50 for EDA and EDE, and 25 for LAC, MECA, and SSA, according to the differentiated carbon prices in Parry et al. 2021 (IMF Staff Climate Note No 2021/001). Finally, the current shares are calculated

[17] Intuitively, this is because developing countries tend to have higher "carbon multipliers". See Economic Activity Indicators | Climate Change Indicators Dashboard (imf.org).

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based on the average of PPP GDP over a three-year period, 2016-2018, at country level with a coverage of 172 (out of 189) member countries in 2020, and are summed up by regions.

III. Green Quotas and Climate Scenarios

A. Evolution of the global economy under climate scenarios

Global temperature has now increased by 1.1 °C since pre-industrial times and is expected to rise further, with large upside uncertainty. This uncertainty is both due to tail risks and the presence of tipping points,[18] and uncertainty about the speed and magnitude of the mitigation efforts. Scientific evidence overwhelmingly attributes climate change to an exponential increase in GHG emissions,[19] with the combustion of fossil fuels contributing to about 80 percent of GHG emissions worldwide.[20] To this date, 193 parties[21] have committed to the Paris agreement aiming to reach net zero global emission, but a large ambition gap remains.[22] The magnitude and frequency of extreme weather events (drought, forest fires, floods, hurricanes, etc.) has increased, compounded the already large expected economic costs of gradual change in climate.

In order to assess the potential climate impact on each country's quota over the next decades, we rely on a variety of scenarios and their projections designed by climate scientists and economists, known as Shared Socioeconomic Pathways (SSPs). While these scenarios share the same scientific underpinning and are widely recognized as internally consistent projections regarding the interdependencies between economic activity, policies, and climate change, they rely on different assumptions on six socio-economic elements, that is demographics, human development, economy and lifestyle, policies and institutions, technology, and environment and natural resources.[23] Each SSP scenario represents a path for potential developments of climate change situations in the next decades. For example, SSP1 is the most benign scenario assuming all countries are fully aware of the importance of adaption and mitigation measures and achieve a green world under coordinated effort. Box 2 provides details on how these scenarios are constructed.

Quota shares calculated using the current quota formula follow different trajectories under the five established climate scenarios. Given that the paths and outcomes of climate change are highly uncertain and policy dependent on certain assumptions of socio-economic factors, instead of establishing ad-hoc scenarios, we use the 5 SSP scenarios at the baseline for our discussion, which do not consider the development of climate policies and thus can be regarded as "no climate policy" baseline scenarios.[24]

Countries' GDP shares indeed show different trends under the SSP projections. Figure 3 below plots the GDP projections for 2035 and 2050 under the 5 SSP scenarios at baseline for the WEO country groups. We have three main observations. First, compared with the current GDP share in the quota formula calculated based on

[18] Tipping points create substantial uncertainty to the downside on the extent of the shock, as large catastrophic events might happen with an unknown probability. See Dietz et al. 2021.

[19] GHG emissions are just one part of the problem. Pollution, biodiversity collapse etc. are other negative externalities associated with economic activities contribution to climate change.

[20] Haberl et al. 2020.

[21] 192 countries plus the European Union.

[22] Masson-Delmotte et al. 2021.

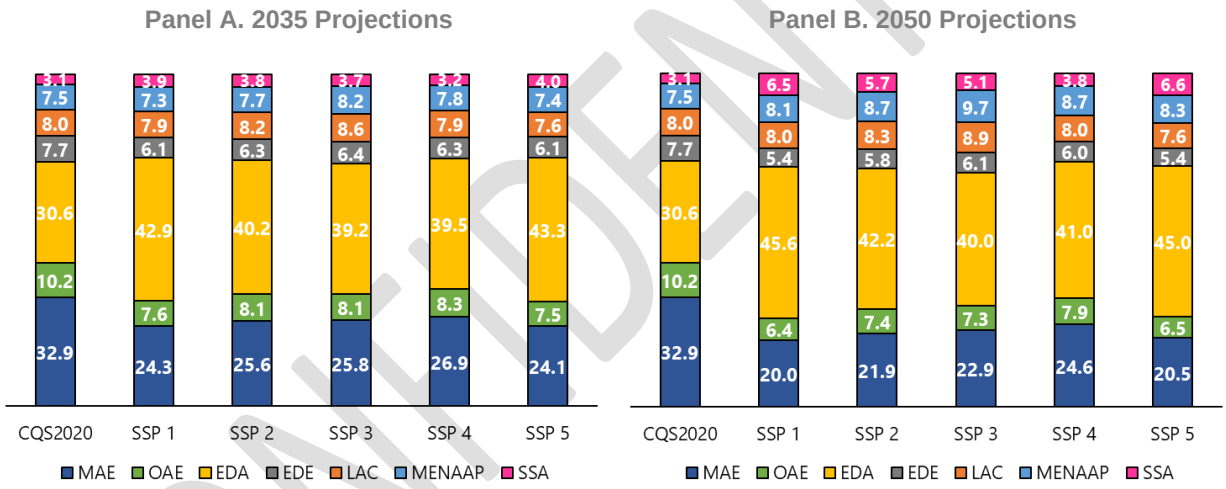
[23] O'Neill et al. 2014.

[24] Dellink et al. 2017.

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the data from Quota Data Update 2020, EDA gains significantly in 30 years from now across all SSPs, whose share of global GDP PPP could amount from 40 percent to 45.6 percent across scenarios, a 50 percent increase from current levels. The gains in EDA come mostly from compressing the shares of AEs, both MAE and OAE. The shares of other country groups remain relatively stable across all SSPs compared with the current global GDP share. The increase in EDA suggests a significant shift of calculated quota share from AEs to EDAs in the way leading up to 2050. Second, SSA benefits more in SSP5, a scenario where countries focus on economic growth regardless of high fossil fuel demand and environmental damages, and in SSP1 with increasing global commons and reducing inequality. Lastly, EDA contributes the most to global GDP share, with EDE, LAC and MENAAP being relatively stable across different SSP scenarios. Note that this has not taken into account any developments in CO2 emissions under the SSP scenarios. As regions fare differently under all scenarios in terms of GDP shares, the current quota formula increases the risk of providing wrong incentives, or failure to act, to safeguard the global common goods in terms of promoting GHG reducing measures and a swifter climate transition, especially for the countries that will experience large share gains.

Figure 3. Comparison of Future Brown Shares of PPP GDP by WEO Classification (in percent) 1/ 2/



Source: IIASA SSP Public Database, IMF staff calculation.

1/ Countries are classified based on WEO October 2021 groupings.

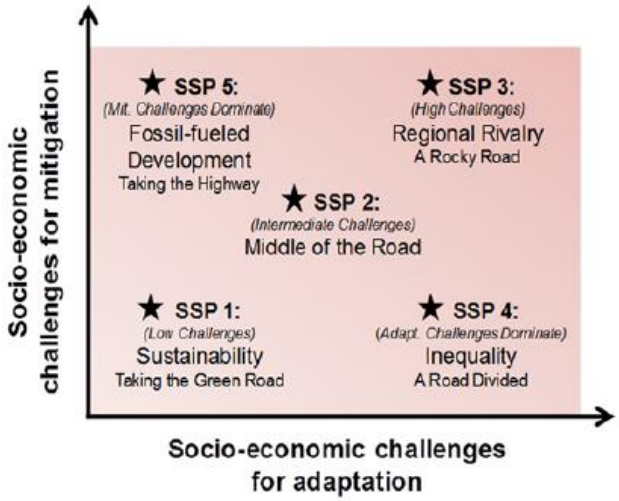
2/ PPP GDP shares in CQS2020 are calculated based on the average of PPP GDP over a three-year period, 2016-2018, at country level with a coverage of 172 (out of 189) member countries in 2020, and are sumed up by regions. The shares of SSP scenarios are based on 2035 and 2050 projections at baseline by the OECD. See Appendix I for details about IMF members included in the SSP database.

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Box 2: Shared Socioeconomic Pathways Scenarios and Representative Concentration Pathways [1/]

Shared Socioeconomic Pathways are narrative based-scenarios developed by climate economists capturing different combinations of mitigation and adaptation. Each scenario starts by a set of demographic, economic, environmental, governance, policy, social, and technological assumptions. The five SSPs can be briefly summarized as follows:[2/]

SSP1 Sustainability – *Taking the Green Road* The world shifts gradually, but pervasively, toward a more sustainable path, emphasizing more inclusive development that respects perceived environmental boundaries. This scenario implies low challenges to mitigation and adaptation due to the strong and effective institutions and international cooperation, the development of technology and environmentally friendly solutions, as well as the improvements in economic development and human well-being.



SSP2 *Middle of the Road* The world follows a path in which social, economic, and technological trends do not shift markedly from historical patterns. In this scenario, the world faces moderate challenges to mitigation and adaptation, because of moderate sustainability and environmentally friendly transition, continued reliance on unconventional energy sources, moderate technological change, moderate international cooperation, uneven development and income inequality within and across countries.

SSP3 *Regional Rivalry – A Rocky Road* A resurgent nationalism, concerns about competitiveness and security, and regional conflicts push countries to increasingly focus on domestic or, at most, regional issues. This scenario shows high challenges to mitigation considering continued fossil fuel demand, high energy intensity, weak global institutions, and low technological change. It also implies high challenges to adaptation because of low investments in human capital and low access to health facilities, water and sanitation that result in vulnerable human and natural systems.

SSP4 Inequality – *A Road Divided* Highly unequal investments in human capital, combined with increasing disparities in economic opportunity and political power, lead to increasing inequalities and stratification both across and within countries. In this scenario, challenges to mitigation are low due to the investment of low carbon energy sources, rapid development in high-tech economies and sectors, and effective capability of a well-integrated political and business class; challenges to adaptation are high considering substantial proportions of populations are at low levels of development, have limited participation in environmental and societal issues, and have low access to education and healthcare.

SSP5 Fossil-fueled Development – *Taking the Highway* This world places increasing faith in competitive markets, innovation and participatory societies to produce rapid technological progress and development of human capital as the path to sustainable development. This scenario implies high challenges to mitigation as the continued and strong dependence on unconventional energy sources, as well as lack of global environmental concerns. It shows low challenges to adaptation because of rapid technological development, high economic growth, and the attainment of development goals.

Each SSP baseline scenarios is complemented by a set of assumption reflecting uncertainties linking GHG emissions and their impact on the climate. Following the IPCC assessment reports, this is accomplished via Representative Concentration Pathways (RCPs), which embeds different radiative forcing levels[25], ranging from 1.9 to 8.5 W/m^2. Together with SSP, RCP allows climate researchers to explore a wide range of possible climate futures.[3/]

1/ Source of the text figure: O'Neill et al. 2014.

2/ Riahi et al 2017 and O'Neill et al. 2014.

3/ See Van Vuuren et al. 2021.

B. Green Quotas across climate scenarios

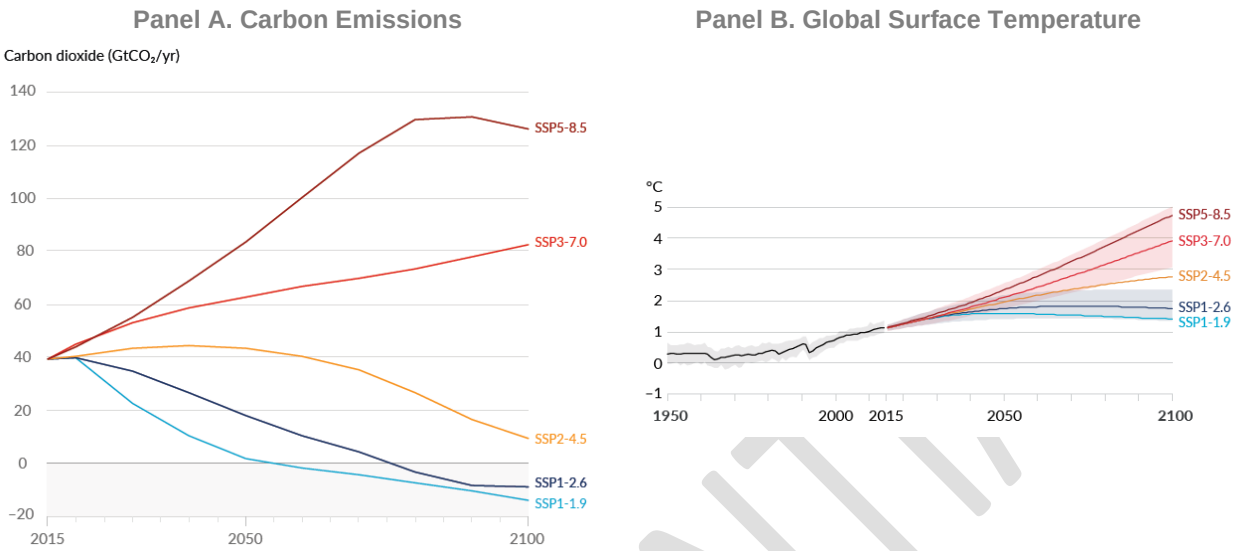
In this section, we analyze how a simple green GDP formula would affect quota shares under SSPs and mitigation scenarios. As introduced in Box 2, each SSP scenarios is complemented by a set of RCPs indicating different possible levels of radiative forcing values in 2100. Five illustrative SSP-RCP scenarios were selected, specifically SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5, aligned with the IPCC selection[26], reflect the development of socioeconomic drivers incorporated with climate forcing targets and associated policy efforts. They include scenarios with rapidly rising carbon emissions, such as SSP3-7.0 and SSP5-8.5 who reach 63 and 83 gigatonnes of CO2 in 2050, respectively, scenario (SSP2-4.5) with relatively unchanged carbon emissions in 2050 compared with current levels, and scenarios (SSP1-1.9 and SSP1-2.6) with strict emission reduction targets and therefore would reach net zero emissions before 2060 and 2080, respectively (Figure 4, Panel A). Figure 5 shows the regional concentration of the carbon emissions in 2050, as Asian countries account for the largest proportion, REF the fourth, and LAM the fifth across scenarios, while OECD and MAF exchange rankings under SSP1-1.9 as the world would shift to a more sustainable path. Only SSP1-1.9 and SSP1-2.6 can fulfill the Paris Agreement's goal to limit global warming to well below 2°C (Figure 4, Panel B).

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[25] Radiative forcing is a scientific concept measuring how much energy flowing into the atmosphere represented by the unit Watt/metre^2. The more positive the radiative forcing value is, the more surplus between the amount of energy earth receives from sunlight and the amount it radiates to space.

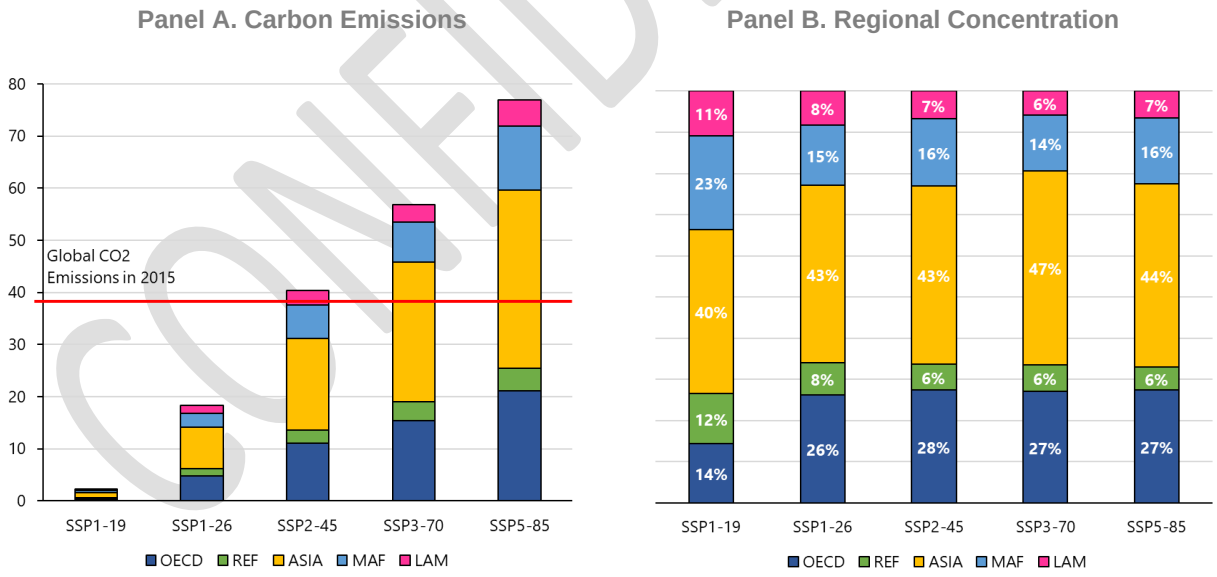
[26] Masson-Delmotte et al 2021. IPCC, 2021: Summary for Policymakers.

Figure 4. Carbon Emissions and Global Surface Temperature Under Five Climate Scenarios



Source: Masson-Delmotte et al 2021. IPCC, 2021: Summary for Policymakers.

Figure 5. Carbon Emissions and Regional Concentration in 2050 (in gigatonnes)



Source: IIASA SSP Public Database, IMF staff calculation.

We implement the simple formula from section II by using GDP and CO_2 emissions projection data from the SSP database under the five illustrative scenarios. Although the SSP database predicts GDP levels by individual countries which enables us to plot Figure 3, it only predicts CO_2 emissions for defined groups of

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countries, namely OECD, REF (Reforming Economies of Eastern Europe and the Former Soviet Union), ASIA, MAF (Middle east and African countries) and LAM (Latin American countries). We show the green GDP calculations for these groups under different SSPs in 2050 in Figure 6. Box 2 explains the treatment of mitigation efforts (through incorporating the RCPs). For carbon pricing, we assume an annual increase of carbon price from 2020 to 2050 by 1.5 USD per tCO_2, referring to the annual price rise of 1.5 USD assumption[27] in Parry et al. 2021 (IMF Working Paper WP/21/236), based on the price levels used in the calculation in Figure 2. The differentiated prices are then calculated as one third of the uniform price for LAM and MAF, and two thirds for ASIA and REF. This approach is aligned with the differentiated price levels in Parry et al. 2021 (IMF Staff Climate Note No 2021/001). Note that in this case the carbon "price" used for quota calculation purposes is just for illustration which does not coincide with the actual carbon price paid by states, and can be altered for other quota simulations.[28] Therefore, the uniform carbon price would be 120 USD per tCO_2, and differentiated prices would be 120 for OECD, 80 for ASIA and REF, and 40 for LAM and MAF (Figure 6, Panel C).

GDP shares are expected to evolve compared to the current ones under different SSPs in 2050. Figure 6 panel A shows the PPP GDP shares without greening the PPP GDP variable for the 5 groups of countries projected by SSP scenarios, in contrast to the previous discussed WEO groupings. Asian countries will gain under all scenarios in 30 years from now, compressing the shares mainly from OECD countries. MAF countries will gain slightly, while LAM and REF countries remain relatively stable.

Replacing PPP GDP with a green PPP GDP does not alter in a major way the balance between regional groups. This is primarily reflected in the fact that in more benign scenarios (SSP1 and SSP2), the green GDP shares are very close to GDP shares across groups (Figure 6, Panel B). One reason for the small adjustment in GDP is that we are not able to subtract other components contributing to negative (local) externalities, as in Stjepanović et al. 2017 approach in calculation due to data limitations, which, after all, are not considered as global externality. The advantage of having small adjustments is that the approach does not initiate abrupt redistributions of shares in the quota calculation.

However, greening quotas would have some redistributive impact. Compared to the case without greening GDP, LAM countries will gain shares under all scenarios and pricing settings, except for SSP1-2.6 under the differentiated price. So do Asian countries except for SSP3-7.0. OECD countries will always be worse off under the differentiated price (left bar in Figure 6, Panel B) compared to the uniform price (right bar in Figure 6, Panel B). MAF countries will always lose absolute shares under the uniform price but will gain shares under the differentiated price. This suggests that, if one of the goals of quota adjustments is to protect shares in developing/poor countries, such goal can be met with prices appropriately designed and these countries can receive slightly higher quotas in the green scenarios than in the brown scenario.

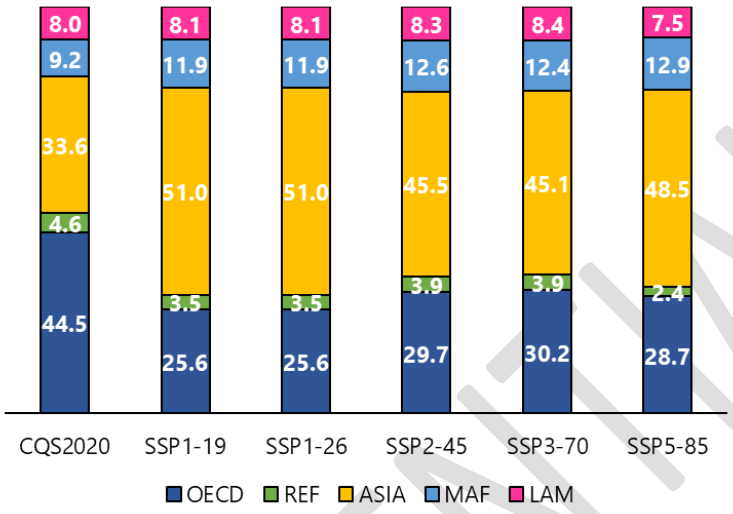
[27] The Parry et al. 2021 (IMF Working Paper WP/21/236) assumed that "prices for intervening for earlier years [2020] are inferred assuming prices rise annually at $1.5 per ton". We replicated the assumption for 2020-2050 for illustration purpose.

[28] Carbon pricing at state or regional level, while growing, still suffer from a low price and incomplete coverage. See More Countries Are Pricing Carbon, but Emissions Are Still Too Cheap – IMF Blog.

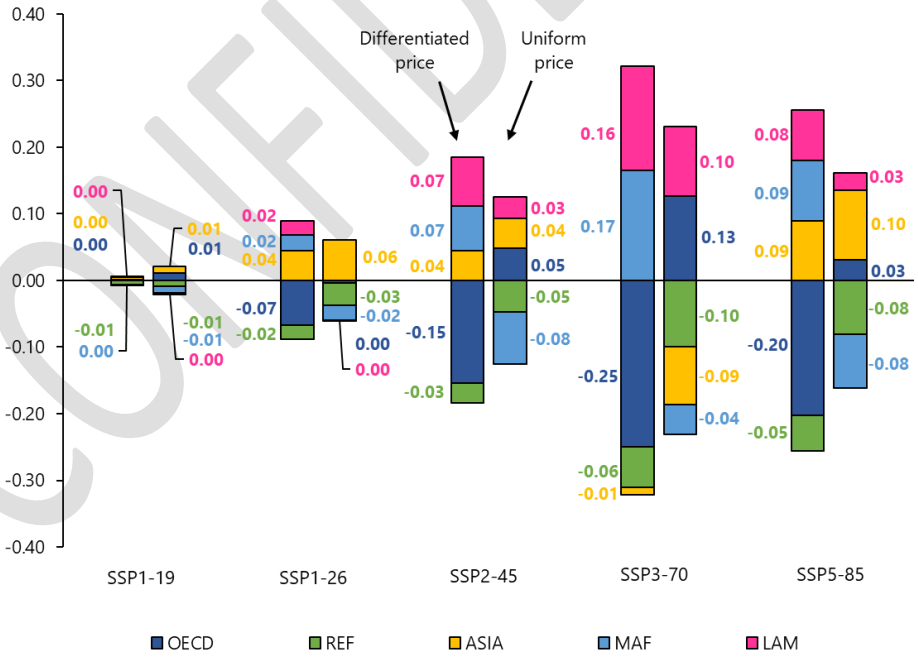
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Figure 6. Comparison of Future Green Shares of PPP GDP by SSP Classification [1/]

Panel A. Without Greening PPP GDP: Current vs. 2050 Projections (in percent) [2/]



Panel B. Greening PPP GDP in 2050: Change of Shares Based on Uniform and Differentiated Carbon Price (in percent) [3/]



Panel C. Presumed Carbon Price in 2050 (in current USD/tCO₂)

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	Uniform Price	Differentiated Price				
		ASIA	LAM	MAF	OECD	REF
SSP1-19	120	80	40	40	120	80
SSP1-26	120	80	40	40	120	80
SSP2-45	120	80	40	40	120	80
SSP3-70	120	80	40	40	120	80
SSP5-85	120	80	40	40	120	80

Source: IIASA SSP Public Database, IMF staff calculation.

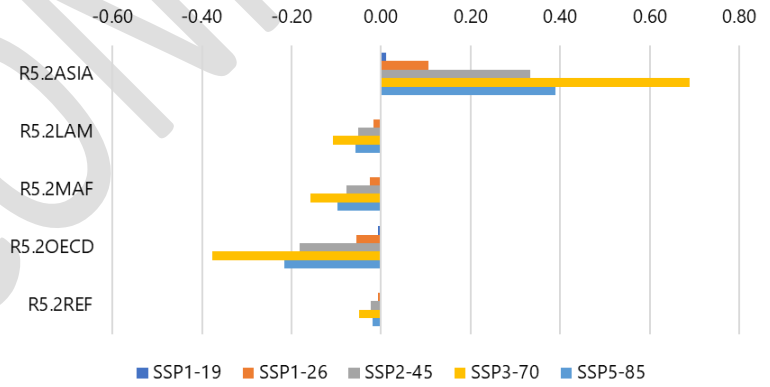
1/ Countries are re-classified based on SSP regional categories: (1) OECD refers to the OECD 90 and EU member states and candidates; (2) REF refers to the Reforming Economies of Eastern Europe and the Former Soviet Union; (3) ASIA includes Asian countries with the exception of the Middle East, Japan and Former Soviet Union states; (4) MAF refers to the countries of the Middle East and Africa; and (5) LAM refers to the countries of Latin America and the Caribbean. See Appendix I for details.

2/ PPP GDP shares for regions shown in the chart in CQS2020 are calculated first on country level based on the average of PPP GDP over a three-year period, 2016-2018, with a coverage of 172 (out of 189) member countries in 2020, and are then summed up by regions. The shares of SSP scenarios are based on 2050 projections at baseline by the OECD. See Appendix I for details about IMF members included in the SSP database.

3/ The Change of PPP GDP shares is calculated as green PPP GDP shares minus brown (non-green) PPP GDP shares. Left bars of each SSP scenario are based on differentiated carbon price, while right bars are based on uniform carbon price. The carbon price is expressed in current USD per tCO2.

Finally, greening quota, would provide incentives for countries to step up efforts even when other countries miss their reduction targets. To see this, we perform a thought experiment in which a group of countries achieves net zero emission while others keep their CO_2 emissions at the level forecasted by the SSP scenario.[29] Suppose, for instance, that Asian countries were to reach zero emissions in 2050 while the rest of the world maintain their emissions at the level forecast by SSP3-7.0, then this would result in a redistribution of about 0.69 of quota shares [see Figure 7]

Figure 7. Redistribution of Shares under emission scenarios



Source: IIASA SSP Public Database, IMF staff calculation.

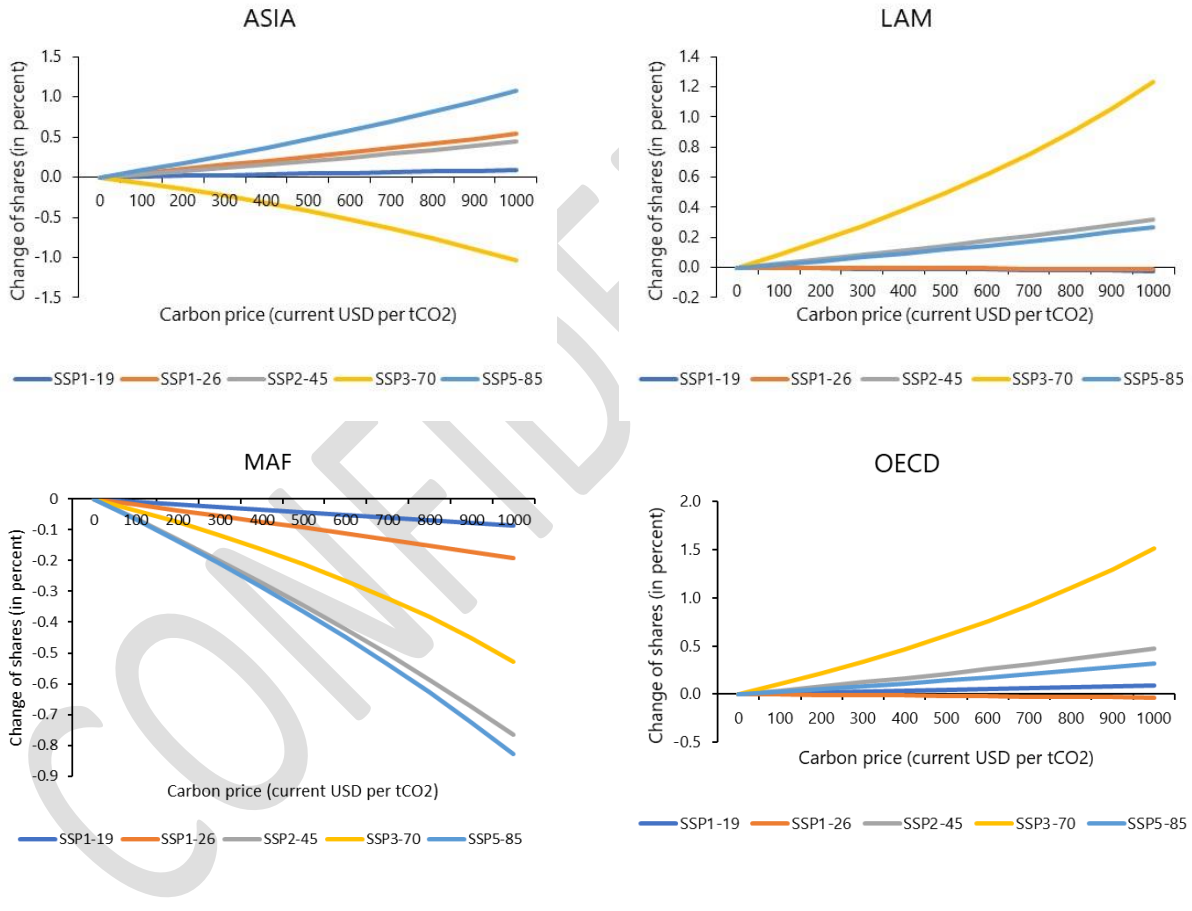
Note:

[29] This calculation is illustrative only and meant to provide a rough order of magnitude of the potential incentive effects. SSPs are conceived as a consistent set of assumptions and deviations on the part of some countries should trigger responses from other countries. This potential response is absent from the partial equilibrium thought experiment conducted here. From this perspective our calculations can be understood as potential upper bounds.

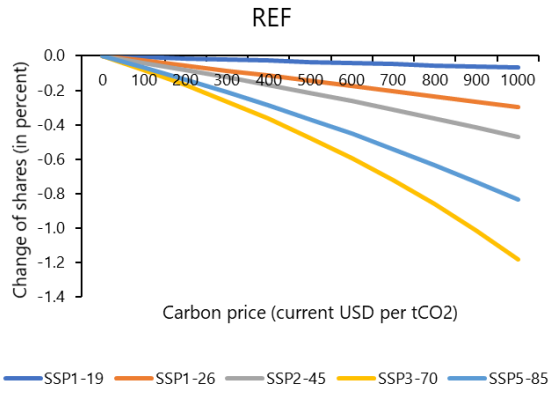
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Rising carbon prices further amplify the patterns described above. Change of shares for each country group is within 2 and 3 percentage points under the uniform and differentiated price settings, respectively. In the case of a uniform price of carbon, as the price rises to $1000 per metric ton, MAF and REF countries will lose shares under all scenarios, while ASIA countries will gain shares except under SSP3-7.0, LAM countries except under SSP1-1.9 and SSP1-2.6, and OECD countries except under SSP1-2.6 (Figure 7). With differentiated prices, the shares of MAF countries will be increasing except under SSP1-1.9 yet those of OECD countries will be decreasing (Figure 8), which emphasize the finding that appropriately-designed prices can help protect the shares of developing and the poorest countries.

Figure 7. Effect of Increasing Uniform Carbon Price on Country Groupings' Changes of Shares in 2050 [1/]



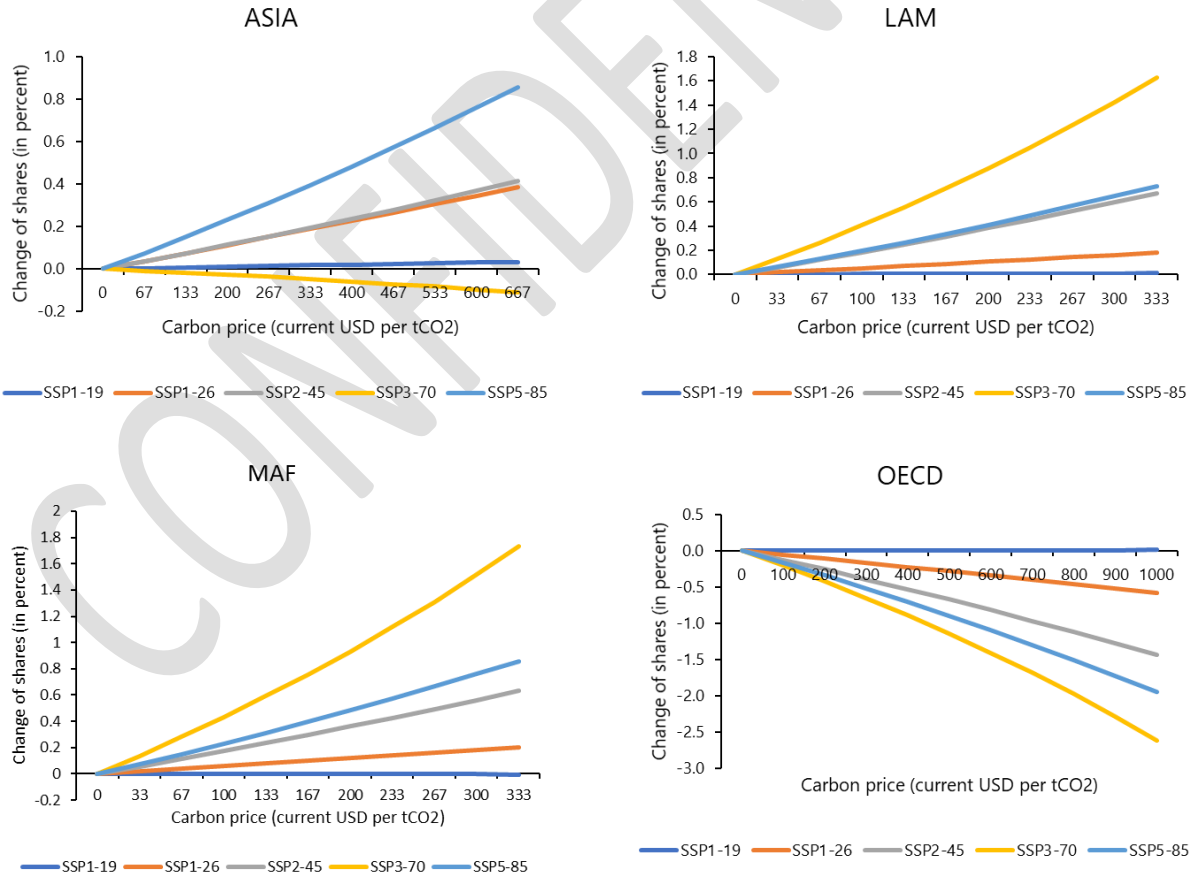
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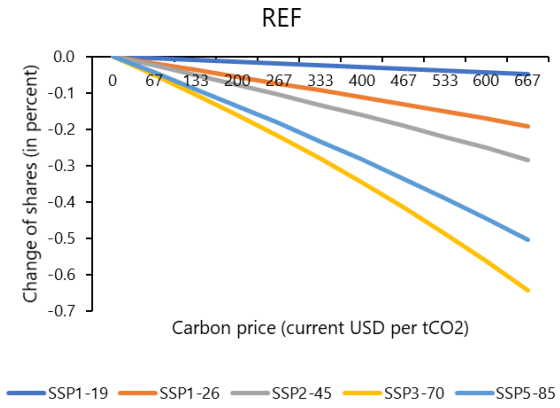
Source: IIASA SSP Public Database, IMF staff calculation.

1/ The change of PPP GDP shares is calculated as green PPP GDP shares minus brown (non-green) PPP GDP shares at a uniform price across country groups, such as at 200 USD per tCO_2.

Figure 8. Effect of Increasing Differentiated Carbon Price on Country Groupings' Changes of Shares in 2050[1/]



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Source: IIASA SSP Public Database, IMF staff calculation.

1/ The change of PPP GDP shares is calculated as green PPP GDP shares minus brown (non-green) PPP GDP shares at differentiated prices that the price for ASIA and REF would be two thirds of that for OECD, and the price for LAM and MAF would be one third of that for OECD. See IMF Staff Climate Note No 2021/001. The differentiated is calculated by keeping constant the ratio between Figure 6 Panel C

IV. Conclusion

The approach presented is in this paper provides a practical method of greening Fund quotas with two advantages: it preserves much of the current quota structure and, under moderate carbon prices, would not generate dramatic reallocations. In addition, it shows that realigning Fund quotas with emissions reductions pledged under the Paris Agreement can be done in a transparent manner, based on widely available data, and which could help reframe incentives for climate policy action.[30]

The proposed method retains a large degree of flexibility. For instance, differentiated carbon prices can be used to protect the representation of the poorest members of the Fund -- a long-standing feature of quota reviews --. Furthermore, by subjecting the variables used to construct our measure of green GDP to the same rigorous data standards as used for the other quota variables, the Fund would be in a position to offer a robust, cross-countries series of Green GDP that could serve as a basis for analyses by many third-party organizations (Civil Society Organizations, other international organizations). That variable could also usefully be featured alongside key economic indicators in IMF country publications. A similar approach to greening the quota formula could be also applied to other quotas-based institutions such as Multilateral Development Banks.[31]

Finally, further refinements of the approach can be considered. Consistent with scientific evidence, a more comprehensive approach would need to capture all GHG emissions, including methane emissions, as well as other positive and negative climate externalities, for instance by taking into account carbon absorbed by ocean and forest cover. A main limitation at the current juncture is the lack of high-quality datasets covering all these variables across the Fund membership. As global work and coordination around climate commitments gather pace, better datasets will become available and allow a more comprehensive greening of Fund quotas.

[30] See Mercure et al. 2021.

[31] The World Bank also relies on a formula using GDP as one of its main variables.

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Annex I. Country Groupings

The five country groupings used in the SSP scenarios are defined in the IIASA database[32] as follows:
ASIA: includes most Asian countries with the exception of the Middle East, Japan and Former Soviet Union states.
LAM: includes the countries of Latin America and the Caribbean.
MAF: includes the countries of the Middle East and Africa.
REF: includes countries from the Reforming Economies of Eastern Europe and the Former Soviet Union.
OECD: includes the OECD 90 and EU member states and candidates.

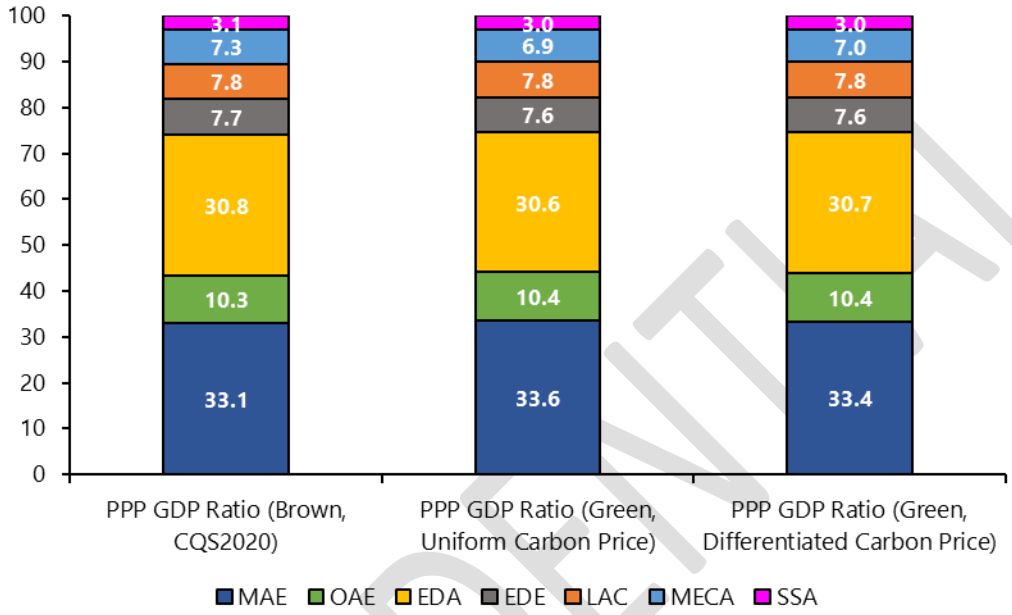
There are 175 countries (and/or regions) in the calculation that can be attributed to both the SSP scenarios and the WEO classification. There are 15 (out of 190) of the IMF members are currently not in the SSP scenarios, namely Andorra, Antigua and Barbuda, Dominica, Grenada, Kiribati, Kosovo, Marshall Islands, Micronesia, Nauru, Palau, San Marino, Seychelles, South Sudan, St. Kitts and Nevis, and Tuvalu.

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[32] SSP Public Database (Version 2.0). https://secure.iiasa.ac.at/web-apps/ene/SspDb/.

Annex II. Additional Calculations

Figure II.1. Comparison of Current Brown and Green Shares of PPP GDP by WEO Classification, Considering Global and Local Externalities (in percent) [1/ 2/]



Source: IMF Finance Department; World Development Indicators database (updated by June 30, 2022), World Bank; Doing Business database (updated by August 18, 2021), World Bank; What A Waste Global database (updated by March 4, 2019), World Bank; and IMF staff calculation.

1/ Countries are classified based on WEO October 2021 groupings.

2/ PPP GDP data are from the Quota Data Update 2020 database. The Green PPP GDP indicator is then calculated based on the formula specified in footnote 11, by subtracting the cost of CO_2 emissions, the cost of natural resource consumption, and the cost of environmental depletion. The CO_2 emissions (from the burning of fossil fuels and the manufacture of cement), gross national income (expressed in PPP term), and the natural resources depletion (% of GNI) data were retrieved from World Bank's World Development Indicators database; the price of electricity data (in US cents) from World Bank's Doing Business database; and the total waste generated data from World Bank's What A Waste Global database, which were collected in different years and from various sources. We assumed the carbon price as 75 USD per tCO_2 for all countries for the three years in calculation (2016-2018), considering the carbon price level as at least 40-80 USD per tCO_2 by 2020 in Stiglitz et al. 2017 and the annual price rise of 1.5 USD prior to 2020 assumption in Parry et al. 2021 (IMF Working Paper WP/21/236), consistent with the Paris temperature target limiting global warming to well below 2°C. We also experimented with differentiated carbon price for different country groups, 75 for MAE and OAE, 50 for EDA and EDE, and 25 for LAC, MECA, and SSA. We assumed that 74 kWh per metric ton of electrical energy would be used for a metric ton of waste, according to Stjepanović et al. 2017. Finally, the current shares are calculated based on the average of PPP GDP over a three-year period, 2016-2018, at country level with a coverage of 172 (out of 189) member countries in 2020, and are summed up by regions.

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